

JAKUB JURKIAN

Java Backend Developer

Gdańsk, Poland

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ABOUT ME

Computer Science student at the University of Gdańsk, building backend systems with **Java** and **Spring Boot**. I care about writing well-tested code I'd be comfortable handing off to someone else: I write key logic test-first and record architectural decisions as ADRs. Seeking a Java Backend Developer or Software Engineering internship - available **now**, part-time alongside my studies - or a full-time junior role from **January 2027**.

TECHNICAL SKILLS

Languages: Java 21+, SQL, Python, TypeScript

Frameworks & Testing: Spring Boot 3+, Spring Security, Hibernate/JPA, Testcontainers, JUnit 5, Mockito

Infrastructure & Databases: PostgreSQL, Redis, Liquibase/Flyway, Docker, GitHub Actions CI/CD, MongoDB

Architecture & Methodologies: RESTful APIs, TDD, Architecture Decision Records (ADRs), Git, GitHub Issues/PR workflow

Also exploring: React, Node.js/Express

PROJECTS

Fleet Management API (VeloCity) | *Java 25, Spring Boot 4, PostgreSQL* | [GitHub](#) | [Demo](#) 2026

- Built a Spring Boot REST API (plus a separate React/TypeScript client) for an e-bike rental platform, secured with a stateless JWT filter chain and a Redis-backed token blacklist for logout.
- Prevented double-booking at the database level with a PostgreSQL exclusion constraint (GiST over date ranges) mapped to HTTP 409, verified by a Testcontainers concurrency test.
- Modelled the reservation lifecycle as a domain state machine with @Version optimistic locking against lost updates; schema versioned with Liquibase migrations.
- Wrote the BigDecimal pricing engine test-first; 40+ unit and integration tests run in GitHub Actions, and a deploy workflow ships the Docker image to an Oracle Cloud VM.

LLM Evaluation API (MockBean) | *Java 21, Spring Boot 3, LangChain4j, pgvector* | [GitHub](#) 2026

- Built a Spring Boot API that grades technical interview answers using an LLM as judge, in RAG (PostgreSQL + pgvector) or plain-prompt mode; Claude, Gemini or a local Llama 3.1 is selected via Spring profiles.
- Benchmarked 3 models × 2 modes on 40 labelled answers with bootstrap confidence intervals (Python): Claude reached 92.5% accuracy, and RAG did not improve accuracy for any model.
- Used Flyway instead of Hibernate auto-DDL, and persisted every evaluation for later analysis of model consistency.

E-Commerce Marketplace API (Grailkits) | *Express.js, Keycloak, PostgreSQL/MongoDB* | [GitHub](#) 2026

- Split a marketplace backend into four services (API Gateway, Catalog, Order, Review), each independently containerized.
- Secured requests using Keycloak (OAuth 2.0 + PKCE); the API Gateway validates JWT tokens and handles rate-limiting with Redis before forwarding traffic to the services.
- Chose PostgreSQL for orders (needed transactions) and MongoDB for product details and reviews (flexible schema).

EDUCATION

University of Gdańsk Gdańsk, Poland
B.Sc. Computer Science (Full-time) 2024 - Present

- **Relevant Coursework:** Algorithms & Data Structures, Industrial Applications, Databases, Concurrent Programming, Web Application Security, Microservices Architecture.
- **Curricular Internship:** The 6th semester (starting January 2027) is entirely cleared of academic classes and fully dedicated to a mandatory professional internship (720 hours) - guaranteeing full-time availability.

IT Technical School (Maciej Rataj ZS) Reszel, Poland
IT Technician Diploma (Specialization: Administration and Programming) 2019 - 2023

LANGUAGES

English: B2+ - daily working language for documentation, code, and technical communication.

Polish: Native.